

Example Library First Principles Driven Hybrid Model for Heat Exchanger Heat Transfer Coefficient (UA) in Aspen Plus

Products: Aspen Hybrid Models for Aspen Plus

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Example Details

The AI Training feature in Aspen Plus enables you to use data and machine learning techniques to improve the accuracy of one or more models; the primary advantage of this approach over traditional data-driven machine learning models is that the training is done within the constraints of consistent physical properties and material and energy balances.

We highlight the different steps required to create such a First Principles Driven Hybrid Model to accurately capture these effects.

- Step 1 – Import Data
- Step 2 – Condition Data
- Step 3 – Evaluate Model
- Step 4 – Build, Train and Validate Model
- Step 5 – Deploy Model



The attached PDF shows the steps to follow to ‘train’ a model with plant data using First Principles-Driven Hybrid Models in Aspen Plus to help us optimize the performance of a Heat Exchanger’s Heat Transfer coefficient (UA).

Keywords

First Principles, Aspen Plus, AI, Train HeatX, UA